TeliAI Campaign Analytics Platform

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Abstract

This report presents a campaign analytics platform built for TeliAI to process and analyze SMS campaign messaging data at scale. The system processed 67K+ conversation threads and 83K+ messages through a LangGraph pipeline, applying LLM-powered sentiment classification to surface regional response rates, optimal engagement timing, and campaign performance insights through an interactive dashboard.

Background and Motivation

SMS campaigns generate large volumes of conversation data, but extracting actionable insights from this data is difficult without the right tooling. TeliAI required a system that could process raw messaging data, reconstruct fragmented conversation threads, apply sentiment analysis at scale, and surface insights through a visual dashboard.

Key metrics targeted for analysis:

- Regional response rates by state and area code
- Engagement timing patterns by day and hour
- Campaign-level sentiment breakdown: positive, neutral, negative
- Campaign-specific performance statistics

Methodology

The system was built around a scalable pipeline connecting raw CSV inputs to an interactive analytics dashboard. Data is loaded into Supabase for structured storage and cross-campaign querying. Python handles the main processing step — cleaning the data, reconstructing conversation threads by joining records on thread ID and sorting by timestamp, and computing metrics including message counts, response rates, and timing patterns.

LLM-powered sentiment analysis is applied through the LangGraph pipeline, with Claude handling classification at each node — labeling each conversation thread as positive, neutral, or negative. Processed results are served through a FastAPI backend and displayed in an interactive dashboard with campaign and region-level filtering.

System Workflow

- Raw CSV data is ingested and loaded into Supabase
- Conversation threads are reconstructed through thread ID joining and timestamp ordering
- LangGraph pipeline routes each thread to Claude for sentiment classification
- Metrics are calculated and served through the FastAPI backend
- Dashboard delivers campaign insights with filtering by campaign, state, and area code

Results

The platform successfully processed 67K+ conversation threads and 83K+ messages from TeliAI's SMS campaign data. An early challenge was identified in the thread reconstruction phase — messages across the raw datasets appeared out of order or lacked sufficient context to form coherent conversations, causing downstream metrics to be miscalculated.

Resolving this required restructuring the transformation process to join records by thread ID and sort by timestamp before passing conversations into the sentiment pipeline. After implementing this fix the pipeline produced reliable and consistent results. The finalized system delivers:

- End-to-end pipeline processing 67K+ threads and 83K+ messages
- LLM-powered sentiment classification with per-campaign breakdowns
- Regional response rate analysis by state and area code
- Interactive dashboard with campaign filtering and engagement timing insights

Conclusion

The platform achieved its goal of transforming raw SMS campaign data into actionable analytics at scale. Thread reconstruction was the most critical processing step, directly enabling accurate sentiment classification and metric calculation downstream. The combination of LangGraph, Claude, FastAPI, and Supabase demonstrated how LLM-powered analysis can be applied effectively to large volumes of real production messaging data.

Future Work

Several directions exist to extend the platform’s capabilities:

- Expand the LangGraph pipeline with multi-step agentic nodes — using Claude to not only classify sentiment but reason over campaign trends and generate natural language insights automatically
- Deploy Claude-powered recommendations that surface the optimal send time, message framing, and target region based on historical campaign performance
- Deploy the system as a fully hosted production application with real-time data ingestion
- Add advanced filtering and cross-campaign comparison views to the dashboard

References

- FastAPI Documentation: https://fastapi.tiangolo.com
- Supabase Documentation: https://supabase.com/docs
- Anthropic Documentation: https://docs.anthropic.com
- LangGraph Documentation: https://langchain-ai.github.io/langgraph